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MIL- PRF-25620D
21 APRIL 1997
SUPERSEDING
MIL-T-25620C
23 JULY 1987

PERFORMANCE SPECIFICATION

TESTER, OXYGEN MASK, HEADSET AND MICROPHONE, TYPE MQ-1A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of portable tester for pre-flight check of pressure demand oxygen masks and personal intercom equipment and EEU-2P goggles. The tester is designated type MQ-1A.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: SA-ALC/TILDD, 485 Quentin Roosevelt Road, Kelly AFB, TX 78241-6524 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC F4128 FSC 6695

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2.2 Government documents

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

BB-O-925	Oxygen, Technical, Gas and Liquid

STANDARDS

FEDERAL

FED-STD-595	Colors Used in Government Procurement

DEPARTMENT OF DEFENSE

MIL-STD-810	Environmental test Methods and Engineering Guidelines.
MS22055	Hose Assemblies-Oxygen Breathing, Connector to

HANDBOOK

DEPARTMENT OF DEFENSE

MIL-HDBK-454	Electronic Equipment, General Guidelines for

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitations.

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AIR FORCE - AERONAUTICAL STANDARDS

AN804	Tee, Flared tube with Bulkhead on Run.
AN924	Nut, Tube, Bulkhead and Universal fitting.
AN929	Cap Assembly, Tube, Pressure Seal.
AN6021	Gage, Panel Mounting Low Pressure Oxygen
AN6029	Indicator, Oxygen Flow

(Unless otherwise indicated, copies of Federal and Military Specifications, Standards, and Handbooks are available from the Defense Automated Printing Service 700 Robbins Avenue, Building 4/D, Philadelphia, PA 19111-5094. ATTN: DoDSSP)

2.3 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM)

ASTM B117	Salt Spray (FOG) Testing

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials,100 Barr Harbor drive, West Conshohocken PA 19428-2959)

(Non-government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.4 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. This specification provides for first article testing. (see 4.3 and 6.4).

3.2.1 Operation. With the tester connected to a 450 pounds per square inch (psi) breathing oxygen cylinder and the communications unit to a 115 VAC outlet, an oxygen mask (see 3.6) shall be tested for leakage while the microphone and headset are being tested for correct operation.

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3.2.2. Oxygen leakage. The oxygen system shall not leak more than 7 psi within one minute under a pressure of 450 psi with source removed. (see 4.5.3)

3.2.3. Load. The case shall withstand an evenly distributed static load of 2.5 psi on all case surface areas without damage. (see 4.5.4)

3.2.4. Water-tightness. The case shall be designed such that water shall not enter the case when tested as specified in 4.5.5.

3.2.5 Salt spray. The case and case accessories (hinges, handle, locks, and reinforcements) shall withstand the salt spray test as specified in 4.5.6.

3.2.6. Handle. The case handle shall be designed and constructed to withstand the test specified in 4.5.7.

3.2.7 Drop. The case shall be able to withstand the free fall drop test specified in 4.5.8.

3.3.1 Components. The tester shall consist, as a minimum, the following components:

Case
Oxygen equipment
  Flow indicator
  Pressure gage
  Oxygen regulator
  Mask hose
  Inlet fitting
Communications and goggle equipment
  Power unit
  Headset and microphone and goggle equipment test circuit

3.3.1.1 Case. The case shall be made of suitable material and finished in a lusterless durable black color no. 37038 of FED-STD-595. The panel shall be recessed so that the accessible components are completely enclosed when the tester is not in use. Case dimensions shall be in accordance with figure 1.

3.3.1.2 Oxygen equipment. The oxygen equipment shall be connected as specified in figure 2.

3.3.1.2.1 Flow indicator. The oxygen flow indicator shall conform to AN6029-1A or be a suitable substitute that meets the same functional and environmental requirements of AN6029-1A and is suitable for use with pure oxygen.

3.3.1.2.2 Pressure gage. The oxygen pressure gage shall conform to AN6021-1B or be a suitable substitute that meets the same functional and environmental requirements of AN6021-1B and is suitable for use with pure oxygen.

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3.3.1.2.3 Oxygen regulator. The oxygen regulator shall conform to MIL-R-6371 or Commercial and Government Entity (CAGE) 99227, Fluid Power P/N 1550 or a suitable substitute that meets the same functional and environmental requirements of MIL-R-6371 and is suitable for use with pure oxygen.

3.3.1.2.4 Mask hose. The regulator end of the hose shall be permanently attached below the panel.

3.3.1.2.5 Inlet fitting. The inlet fitting shall be a T-fitting. The fitting shall be attached to the panel of the tester with a check-nut conforming to AN924D5. A cap conforming to AN929 shall be provided for each of the two inlets. The cap shall be attached to the case by a suitable retaining device.

3.3.1.3 Selection of Electronic Components. Selection of electronic components shall be in accordance with the guidelines set forth in MIL-HDBK-454 corresponding to each electronic component used in the tester.

3.3.1.3.1 Communications and goggle equipment. Communications equipment shall be provided to allow a "talk-out" check of headsets and mask microphones currently in use. The circuitry shall provide clear sound during the "talk-out" check, which is free from any noise or hum that would interfere with the operator's ability to test the communications equipment. The equipment shall consist of necessary circuitry for headset, microphones (carbon and dynamic) and functionally test EEU-2P goggles.

3.3.1.3.2 Power unit. A means to transform 115V, 60 Hz to 28 VDC at 200 mA shall be provided. As a minimum the power unit will include; isolation transformer, "ON" "OFF" switch, power on light, power conditioner, 6 ft heavy duty rubber electrical cord.

3.3.1.3.3 Fuse. The test set shall have circuit protection. A fuse shall be located between the input power and the internal circuit.

3.3.2 Control panel. All electrical connections and control components shall be located on the top panel of the tester. Arrangement and location of the connectors, control knobs, indicator, gage and accessories shall be in accordance with MIL-HDBK-454 Guideline 28.

3.4 Maintainability. The equipment shall be designed as to facilitate maintenance. It shall require a minimum number and types of tools and test equipment to perform maintenance. The tester shall provide optimum accessibility and replaceability of all components. Maintenance procedures shall be provided with the equipment.

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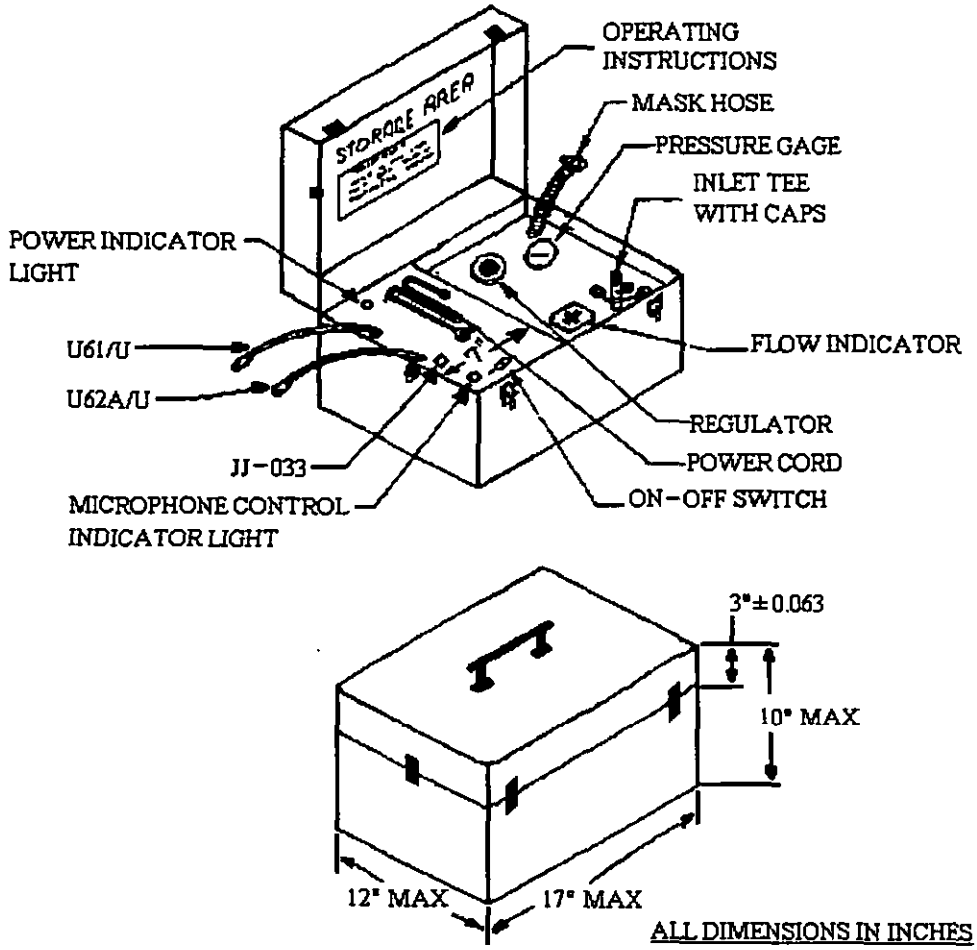


Figure 1. Tester

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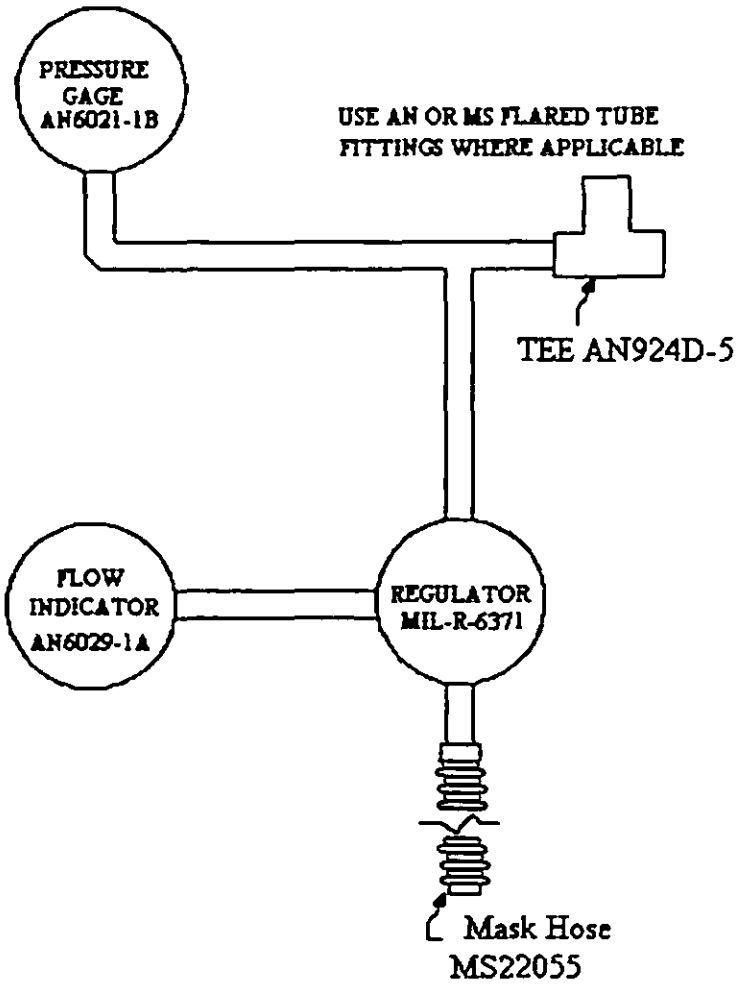


FIGURE 2. Oxygen Equipment

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3.5 Operating instructions. Instructions for operating the tester shall be permanently attached to the inside of the case cover as shown on figure 1. The instructions shall be as follows:

Preliminary instructions.

a. Connect external oxygen supply to one inlet of the test unit with the other inlet capped (450 psi).

b. Set the oxygen regulator to "normal" oxygen position.

c. Open cylinder valve. Allow pressures up to 450 psi (maximum).

d. Plug input into 115V, 60 Hz power source, and turn "ON" - "OFF" switch to "ON" position.

Operating instructions.

a. Don headgear and mask.

b. Set regulator to "normal" position.

c. Connect mask delivery tube to mask hose.

d. Turn regulator to 43M (M=1000).

e. Exhale slowly.

(1) If blinker opens - no leaks.

(2) If blinker remains closed - mask leaks and needs adjustment.

f. Turn regulator to 45M.

g. Exhale slowly.

(1) If no resistance to exhalation, inhalation valve is satisfactory.

(2) If resistance to exhalation, inhalation valve needs cleaning or replacing in accordance with applicable technical order.

h. Plug in headset and microphone. Headset and microphone are functioning properly when the voice is transmitted from the microphone to the headset through the amplifier. (NOTE: When a hand-held microphone is tested, the handle switch must be pressed to talk. The microphone control indicator light should light up when the microphone switch is depressed).

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i. Plug EEU-2P goggles into goggle power cord after donning the helmet. Pull the goggles forward and downward to engage the side latches. The lenses should immediately change to the open, transparent mode.

(1) Wait 10 seconds, then press the press-to-test (PTT) actuator (located above the left front lens), the goggles should rapidly darken to the closed mode. They should rapidly return to the open mode when the PTT Actuator is released.

(2) Use a fast strobe light (SDU-5/E) to illuminate the forward side of the goggles. The goggles should rapidly darken to the closed mode, and then return to the open mode.

(3) Raise the goggles to remove electrical power, the goggle should darken to the closed mode in approximately five seconds.

j. Set oxygen regulator back to "normal" and disconnect the mask, microphone-headset and EEU-2P goggles from the test unit.

3.6 Government loaned property. The Government will loan the following equipment to the contractor upon his request (see 6.5).

ITEM	NSN	QTY
Mask	1660-01-073-7596	1
Goggles	8475-01-017-4473	1
Cable	8475-01-081-1349	1
Cable	8475-01-081-1350	1
Top latch Assy	8475-01-098-0881	1
Cord Assy	5995-01-006-1836	1
Microphone	5965-00-843-9957	1
Headset Assy	5965-00-069-2723	1

EEU-2P Operational Manual will be provided, upon request to contractor for reference material.

3.7 Materials. Nonmagnetic and nonferrous materials shall be used for all parts of the indicator, except where such materials are essential. Metals shall be corrosion resistant or suitably treated to resist corrosion. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with one another.

3.7.1 Protective treatment. When materials are used in the indicator that will deteriorate when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip or scale with age or the extremes of climatic and environmental conditions shall be avoided.

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3.7.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. All components of this tester shall be new and unused.

3.8 Identification plate. An identification plate shall be provided on the rear of the tester case. Information provided shall include the government designation for the tester, manufacturer's part number and the manufacturer's CAGE Code, Serial Number, and government contract number.

3.9 Workmanship. The tester shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, welding, riveting, brazing, plating and painting. All parts shall be free from burrs and sharp edges.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.1.1 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.2.

4.2 Test conditions.

4.2.1 Oxygen. Oxygen used for test purposes shall conform to Type I, grade A of BB-0-925.

4.2.2 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches at mercury and at a temperature of 77 degrees Fahrenheit (25 degrees Celsius) plus or minus 18°F (±7°C), and at a relative humidity of 80 percent or less.

4.3 First article. The first article inspection shall consist of all tests described under Test Methods (see 4.5).

4.3.1 First article test sample. When a first article inspection is required, the item to be tested shall be as specified in paragraph 4.5 Test Methods. The first article test sample shall be identical in all respects to the test sets that are produced as production items.

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4.3.2 First article test report. When specified in contract or order, upon completion of the first article inspection and tests, a test report shall be prepared in accordance with contractor format. The report shall include the results of the tests conducted under examination (see 4.5.1) and operation (see 4.5.2) as well as checklists used during the examination and operation.

4.4 Conformance inspection. Inspection of each production item shall consist of examination (see 4.5.1), operation (see 4.5.2) and oxygen leakage (see 4.5.3).

4.5 Test methods.

4.5.1 Examination. Each tester shall be examined to determine conformance with the requirements of this specification with respect to general construction, finish, workmanship, and marking. One tester in each lot shall be examined to verify conformance of selection of electronic components (see 3.3.1.3).

4.5.2 Operation. The operating instructions listed in 3.5 shall be accomplished on each tester. Prior to plugging in EEU-2P goggles and checking for proper voltage and current output, the following tests should be performed on the power conditioner.

a. With 32 VDC ±1 to input and the output short circuited, the output current shall be 70 mA (min.) to 100 mA (max.).

b. With 28 VDC ±4 to input and a load of 20 mA ±1 using 1400 ohm resistor to output, the output voltage shall be 27V (min).

4.5.3 Oxygen leakage. Rate of leakage shall be determined by applying 450 psi pressure to the tester with the regulator in "normal" position. The rate of leakage shall be determined by pressure drop within the system. Testers which leak more than 7 psi liter per minute when the source is removed shall be rejected.

4.5.4 Load. Any cracking, or permanent deformation or damage to the case, lock, or hinges caused by this load shall be cause for rejection.

4.5.5 Water-tightness. The case shall be rain-tested in accordance with procedure I, "RAIN TESTS" of MIL-STD-810, Method 506.

4.5.6 Salt spray. The case and case accessories (hinges, handle, locks, and reinforcements) shall be subjected to a 30-hour salt spray test in accordance with ASTM B117, at 5 percent salt concentration. Any evidence of corrosion in the case accessories; or any distortion of dimensions, or displacement of fittings of more than 0.067 inch (1.7 millimeter) of the case subsequent to drying, shall be cause for rejection.

4.5.7 Handle. Any empty case shall be loaded with dead weight equal to five times the weight of the complete tester to give even interior loading. The case shall be lifted and rotated through 360 degrees 10 times. Cause for rejection shall be any residual distortion at completion of the lifting test which exceeds 0.067 inch in linear and angular dimensions or linear or angular

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dimensions from the original design specifications. Opening or springing of the case locks, hinges, reinforcements, handle or any permanent distortion of the case during the lifting tests shall be cause for rejection.

4.5.8 Drop. The case shall be subjected to 18 inches free fall drop on a concrete surface, on each of eight corners. A dummy load (15 pounds ±1 pound) shall be used to simulate the weight of the equipment to be installed in the case. Use of the case shall not be impaired after this test.

5. PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory).

6.1 Intended use. The tester covered by this specification is intended for use in determining the proper operation of the dilutor demand, pressure breathing oxygen masks, communication systems, and EEU-2P goggles.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2)
c. Packaging requirements (see 5.1).

6.2.2 Associated Data Item Descriptions (DIDs). This specification is cited in DoD 5010.12-l, Acquisition Management Systems and Data Requirements Control List (AMSDL), as the source document for the following DID. When it is necessary to obtain the data, the applicable DID must be listed on the Contract Data Requirements List (DD Form 1423), except where the DoD Federal Acquisition Regulation Supplement exempts the requirement for a DD Form 1423.

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Paragraph No.	Data requirement title	Applicable DID No.
4.3.2	Test/Inspection Reports	DI-NDTI-80809A

The above DID was current as of the date of this specification. The current issue of the AMSDL must be researched to ensure that only current and approved DIDs are cited on the DD Form 1423.

6.3 First article. When a first article inspection is required, the item should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract, and bidder's part must be in good standing at time of contract award.

6.4 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.6, upon request of contractor.

6.5. Subject term (key word) listing.

Oxygen mask tester
Portable oxygen mask tester
Pre-flight tester, oxygen mask
Tester, oxygen, mask
Tester, portable, oxygen mask
Tester, pressure demand, oxygen mask
Tester, oxygen mask, headset and microphone, type MQ-1A

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Air Force - 99
Reviewer:
DLA - GS

Preparing Activity:
Air Force-82
Agent:
Air Force - 99

(Project 6695-0120)

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INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and
3. The preparing activity must provide a reply within 30 days from receipt of the

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification requirements on current contracts. Comments submitted on this form do not constitute or imply waiver any portion of the referenced document(s) or to amend contractual

I RECOMMEND A CHANGE:	1. DOCUMENT MIL-PFR-25620D	2. DOCUMENT DATE

3. DOCUMENT
TESTER, OXYGEN MASK, HEADSET AND MICROPHONE, TYPE MQ-1A

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	e. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME SA-ALC/TILDD	b. TELEPHONE *(Include Area Code)* (1) Commercial: (210) 925-6314 (2) AUTOVON: 945-6314
c. ADDRESS *(Include Zip Code)* 485 Quentin Roosevelt Rd. Kelly AFB, TX 78241-6424	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041- Telephone (703) 756-2340 AUTOVON 289-

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